

CurrentModule = KiteUtils

Heading Angle

The **heading angle** ψ describes the direction the nose of the kite is pointing, projected onto the plane tangential to the half-sphere at the kite's current position. This projection plane is the **Small Earth (SE) reference frame**.

Coordinate frames involved

Frame	Abbrev.	Convention
Kite Sensor	KS	x : trailing $\rightarrow$ leading edge; y : right wing; z : down
Earth Xsens (NED)	EX	x : North; y : East; z : Down
Earth Groundstation	EG	x : North; y : West; z : Up
Wind	W	x : upwind; y : downwind; z : Up
Small Earth	SE	tangential to unit sphere at kite's position

Transformation chain

The heading vector starts as the unit vector along the kite's nose in the **KS** frame,

$$\mathbf{h}^{\text{KS}} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix},$$

and is then transformed through four frames until it reaches **SE**, where the heading angle is read off.

Step 1 — KS $\rightarrow$ EX (NED): apply Euler angles

Given the kite orientation as Euler angles $(\phi, \theta, \psi_{\text{NED}})$ (roll, pitch, yaw) defined with respect to the NED reference frame:

$$R_{\text{roll}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi & \cos \phi \end{pmatrix}, \quad R_{\text{pitch}} = \begin{pmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{pmatrix}, \quad R_{\text{yaw}} = \begin{pmatrix} \cos \psi_{\text{NED}} & -\sin \psi_{\text{NED}} & 0 \\ \sin \psi_{\text{NED}} & \cos \psi_{\text{NED}} & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

The combined rotation applied in yaw $\rightarrow$ pitch $\rightarrow$ roll order gives

$$\mathbf{h}^{\text{EX}} = R_{\text{yaw}} R_{\text{pitch}} R_{\text{roll}} \mathbf{h}^{\text{KS}}.$$

Step 2 — EX → EG: NED to North-West-Up

The EG frame shares the North axis (x) with NED but flips both y (East → West) and z (Down → Up):

$$R_{\text{EX} \rightarrow \text{EG}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad \mathbf{h}^{\text{EG}} = R_{\text{EX} \rightarrow \text{EG}} \mathbf{h}^{\text{EX}}.$$

Step 3 — EG → W: rotate into Wind frame

Let α be the **down-wind direction** (the direction the wind is blowing *towards*), measured in the EG frame as an angle from North, positive clockwise from above. It is related to the configurable `upwind_dir` by $\alpha = \text{upwind_dir} + \pi$.

$$R_{\text{EG} \rightarrow \text{W}}(\alpha) = \begin{pmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{h}^{\text{W}} = R_{\text{EG} \rightarrow \text{W}}(\alpha) \mathbf{h}^{\text{EG}}.$$

Step 4 — W → SE: project onto the tangential sphere plane

Given the kite's **elevation angle** β and **azimuth angle** φ (in the wind reference frame), the transformation to the Small Earth frame consists of three successive rotations:

$$R_{\text{first}} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \quad R_{\text{az}}(\varphi) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \varphi & \sin \varphi \\ 0 & -\sin \varphi & \cos \varphi \end{pmatrix}, \quad R_{\text{el}}(\beta) = \begin{pmatrix} \cos \beta & 0 & \sin \beta \\ 0 & 1 & 0 \\ -\sin \beta & 0 & \cos \beta \end{pmatrix}$$

$$\mathbf{h}^{\text{SE}} = R_{\text{el}}(\beta) R_{\text{az}}(\varphi) R_{\text{first}} \mathbf{h}^{\text{W}}.$$

Heading angle definition

The heading angle is the angle of the projected nose vector in the SE plane, measured from the x_{SE} axis (pointing towards zenith at the kite's position):

$$\boxed{\psi = \text{atan2}(h_y^{\text{SE}}, h_x^{\text{SE}})}$$

By default (`respos = true`) the result is wrapped to $[0, 2\pi)$; with `respos = false` it is returned in $(-\pi, \pi]$.

A heading of $\psi = 0$ means the kite's nose points towards the zenith direction in the SE plane (the kite is flying “upward”). A heading of $\psi = \pi/2$ means the kite's nose points to the right (in the SE plane, clockwise when viewed from the ground station).

API

```
calc_heading  
calc_heading_w  
calc_clock_angle  
calc_course
```